Independent Research Portfolio Project

Quaternion-Based Attitude Dynamics and Control Simulation for a CubeSat

A 3U CubeSat Rigid-Body Attitude Determination and Control Study

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Abstract

Accurate three-axis attitude determination and control is a foundational requirement for nearly all spacecraft missions, and it becomes especially demanding for CubeSats, whose small size, limited actuation authority, and tight power budgets constrain the achievable pointing performance. This report presents the development and analysis of a quaternion-based attitude dynamics and control simulation for a 3U CubeSat equipped with a three-axis reaction wheel assembly. The spacecraft attitude kinematics are propagated using unit quaternions to avoid the singularities inherent to three-parameter Euler angle representations, while the rotational dynamics are modeled using Euler's rigid-body equations of motion about the spacecraft's principal axes. A proportional-derivative (PD) control law is formulated directly on the quaternion attitude error and body angular velocity to command reaction wheel torques, and is shown analytically, via a Lyapunov argument, to be almost-globally asymptotically stable. The complete model is implemented in MATLAB using a fourth-order Runge-Kutta numerical integrator and exercised over a representative post-deployment scenario combining an initial 19.7°/s tumble with a 120° large-angle slew to a target inertial attitude. Starting from this combined initial condition, the closed-loop simulation converges to within 0.11° of the target attitude and 0.01°/s of zero angular rate within 74.1 s, with the reaction wheels briefly saturating at their 1 mN m torque limit for the first 5.6 s. The results are validated against three independent checks: conservation of the quaternion unit-norm constraint (drift held at floating-point round-off level), agreement between the primary fixed-step integrator and an independent adaptive solver (maximum discrepancy 1.3×10^{-4} degrees), and monotonic decrease of the derived Lyapunov function, confirming the simulated closed-loop behavior matches its theoretical stability certificate. Simplifying assumptions – including a rigid-body spacecraft, idealized torque-source actuators, and the exclusion of environmental disturbance torques – are stated explicitly, and their impact on the fidelity of the results is discussed.

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Nomenclature

$\boldsymbol{q}$	attitude quaternion, $\boldsymbol{q} = [\boldsymbol{\varepsilon}^{\mathsf{T}},\ \eta]^{\mathsf{T}}$
$\boldsymbol{\varepsilon}$	vector (imaginary) part of a quaternion
η	scalar (real) part of a quaternion
$\boldsymbol{q}_e$	attitude error quaternion
$\boldsymbol{\omega}$	spacecraft angular velocity, expressed in the body frame [rad/s]
$\boldsymbol{\tau}$	external/control torque vector, body frame [N·m]
$\mathbf{J}$	spacecraft inertia tensor, body frame [kg·m^2]
$\mathcal{F}_I$	Earth-Centered Inertial reference frame
$\mathcal{F}_B$	spacecraft body-fixed reference frame
K_p	proportional control gain
K_d	derivative (rate damping) control gain
$\mathbf{C}(\boldsymbol{q})$	direction cosine matrix parameterized by $\boldsymbol{q}$
	skew-symmetric cross-product matrix operator

1 Introduction

1.1 Motivation for Spacecraft Attitude Control

Nearly every function a spacecraft performs depends, directly or indirectly, on knowing and controlling its orientation. Earth observation payloads must be pointed at a ground target to within the instrument's field of view; communication antennas must be aligned with a ground station or relay satellite to close a link budget; solar arrays must be oriented toward the Sun to meet the power budget; and propulsive maneuvers require the thrust vector to be pointed along a specific inertial direction to achieve the intended orbit change. The subsystem responsible for meeting these requirements – determining the current orientation from sensor data and commanding actuators to drive the orientation toward a desired reference – is the Attitude Determination and Control System (ADCS) [1, 2].

Attitude control is fundamentally a rotational rigid-body dynamics and control problem: given a mass distribution (the inertia tensor), an initial angular state, and a set of available torques, drive the orientation and angular velocity of the body toward a target state while respecting actuator limits and disturbance environment. This report focuses on the dynamics and control side of that problem – deriving the equations of motion, designing a feedback control law, and validating the resulting closed-loop behavior in simulation – for a small satellite platform where these problems are both highly relevant and tightly constrained.

1.2 Importance of CubeSat Missions

The CubeSat standard, originated at Stanford University and California Polytechnic State University in 1999, defines spacecraft in discrete units ("U") of approximately

2 Spacecraft Mathematical Model

This section develops the mathematical model used to propagate the spacecraft's orientation and angular velocity in time. Two reference frames are defined in Sec. 2.1. The attitude representation problem is discussed in Sec. 2.2, motivating the choice of the unit quaternion over Euler angles. The quaternion kinematic differential equation is then derived in Sec. 2.3, and the rigid-body rotational equations of motion (Euler's equations) are derived in Sec. 2.4. Together, Secs. 2.3 and 2.4 form the coupled state-space model that is integrated numerically in Sec. 5.

2.1 Coordinate Reference Frames

Two right-handed, orthonormal reference frames are used throughout this report.

Earth-Centered Inertial (ECI) frame, $\mathcal{F}_I$. The ECI frame has its origin at the Earth's center of mass. Its $\hat{Z}_I$ axis is aligned with the Earth's mean rotation axis (celestial pole), its $\hat{X}_I$ axis points toward the vernal equinox direction at a reference epoch, and $\hat{Y}_I$ completes the right-handed triad. Unlike an Earth-fixed frame, $\mathcal{F}_I$ does not rotate with the Earth, which makes it (to the accuracy needed here) a valid Newtonian inertial frame in which Euler's rotational equation of motion, derived in Sec. 2.4, holds without the fictitious torque terms that would otherwise arise from a rotating or accelerating reference [3, 1]. All target/reference attitudes in this report are specified as fixed orientations relative to $\mathcal{F}_I$.

Spacecraft body frame, $\mathcal{F}_B$. The body frame is fixed to the spacecraft structure, with its origin at the spacecraft center of mass. Its axes $\hat{x}_B, \hat{y}_B, \hat{z}_B$ are chosen, where possible, to coincide with the spacecraft's principal axes of inertia (Sec. 3), which is the standard convention because it eliminates the off-diagonal products of inertia from the equations of motion in Sec. 2.4. For the 3U CubeSat considered in this report, $\hat{z}_B$ is taken along the long (30 cm) edge of the spacecraft and $\hat{x}_B, \hat{y}_B$ span the square (10 cm × 10 cm) cross-section.

Attitude as a frame transformation. The spacecraft attitude is the orientation of $\mathcal{F}_B$ relative to $\mathcal{F}_I$, represented by the direction cosine matrix (DCM) $\mathbf{C}$ such that a vector $\boldsymbol{v}$ expressed in inertial coordinates is expressed in body coordinates as

$$\boldsymbol{v}_B = \mathbf{C}\,\boldsymbol{v}_I. \tag{2.1}$$

$\mathbf{C}$ is proper orthogonal ($\mathbf{C}^{\mathsf{T}}\mathbf{C} = \mathbf{I}_3$, $\det \mathbf{C} = 1$) and therefore has only three independent degrees of freedom, even though it contains nine entries. Sec. 2.2 discusses two ways of parameterizing $\mathbf{C}$ with a minimal (three-parameter) and a redundant (four-parameter) set, respectively.

Scope note: because attitude determination and Earth-relative pointing laws (e.g., nadir- or Sun-pointing, which require an orbit/LVLH frame) are outside the scope defined in **??**, the target attitude used in Sec. 5 is a fixed quaternion in $\mathcal{F}_I$ rather than a time-varying, orbit-dependent target. This is a simplification relative to an operational pointing law and is revisited in Sec. 7.

2.2 Attitude Representation

2.2.1 Euler angles and the gimbal lock singularity

A common minimal parameterization of $\mathbf{C}$ uses three sequential rotation angles (Euler or Tait-Bryan angles). For the aerospace-standard 3-2-1 (yaw-pitch-roll) sequence, the body frame is reached from the inertial frame by a yaw rotation ψ about $\hat{Z}$, followed by a pitch rotation θ about the new $\hat{Y}$, followed by a roll rotation ϕ about the new $\hat{X}$, so that $\mathbf{C} = \mathbf{C}_1(\phi)\,\mathbf{C}_2(\theta)\,\mathbf{C}_3(\psi)$. The appeal of this representation is purely physical: each of the three angles corresponds directly to an intuitive rotation.

The drawback is that the kinematic differential equations relating the Euler angle rates to the body angular velocity $\boldsymbol{\omega} = [p \;\; q \;\; r]^\mathsf{T}$ contain a $\sec\theta$ term [2, 4]:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \tag{2.2}$$

At $\theta = \pm 90°$, $\sec\theta \to \infty$: the roll and yaw axes become instantaneously aligned (the physical origin of the term *gimbal lock*), one row of Eq. (2.2) becomes undefined, and a finite body rotation rate can require an infinite Euler angle rate to represent it. The spacecraft itself loses no physical degree of freedom at this orientation – only the three-parameter *representation* of its attitude becomes singular. For a spacecraft restricted to small pointing excursions about a single known orientation, the singularity can often be steered around by choosing the reference orientation appropriately. That workaround is not available here: Sec. 5 defines a large-angle slew scenario in which the attitude is not confined to a small neighborhood of any single orientation, so a singularity-free representation is required.

2.2.2 Quaternion representation

The unit quaternion represents an attitude with four parameters instead of three. Writing the quaternion as

$$\boldsymbol{q} = \begin{bmatrix} \boldsymbol{\varepsilon} \\ \eta \end{bmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \in \mathbb{R}^3, \quad \eta \in \mathbb{R}, \tag{2.3}$$

the vector part $\boldsymbol{\varepsilon}$ and scalar part η are related to the Euler axis/angle representation of the same rotation – a single rotation by angle ϕ about a fixed unit axis $\hat{\boldsymbol{e}}$ (Euler's rotation theorem guarantees such an axis always exists) – by

$$\boldsymbol{\varepsilon} = \hat{\boldsymbol{e}} \sin\left(\frac{\phi}{2}\right), \qquad \eta = \cos\left(\frac{\phi}{2}\right). \tag{2.4}$$

No trigonometric inverse of the form $1/\cos\theta$ appears anywhere in the quaternion kinematics derived in Sec. 2.3, so Eq. (2.4) is well-defined for every physical orientation, including $\theta = \pm 90°$ in the Euler angle sense. This comes at the cost of a fourth, redundant parameter: Eq. (2.4) shows $q_1^2 + q_2^2 + q_3^2 + \eta^2 = \sin^2(\phi/2) + \cos^2(\phi/2) = 1$, so only three of the four quaternion components are independent.

The corresponding DCM, obtained by substituting Eq. (2.4) into the Euler axis/angle rotation formula (Rodrigues' formula), is

$$\mathbf{C}(\boldsymbol{q}) = (\eta^2 - \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon})\,\mathbf{I}_3 + 2\,\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\mathsf{T} - 2\eta\,\boldsymbol{\varepsilon}mu,$$

$eq : quat_t o_d cm(2.5)$ where $\boldsymbol{\varepsilon}muistheskew - symmetricmatrixsuchthat\boldsymbol{\varepsilon}mu\boldsymbol{x} = \boldsymbol{\varepsilon} \times \boldsymbol{x}$ for any

$\boldsymbol{x} \in \mathbb{R}^3$. **??** is used whenever the quaternion state needs to be converted to a physically interpretable rotation, such as for plotting an equivalent Euler angle time history in Sec. 6.

2.2.3 Quaternion normalization

Because only a unit-norm quaternion corresponds to a valid, proper-orthogonal DCM through **??**, the constraint

$$\|\boldsymbol{q}\| = \sqrt{\boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} + \eta^2} = 1 \tag{2.6}$$

must hold at all times. Sec. 2.3 shows that this constraint is preserved exactly by the continuous-time kinematic differential equation; however, a discrete-time numerical integrator (Sec. 5) satisfies Eq. (2.6) only approximately, and small errors accumulate over many integration steps. This is monitored directly in Sec. 7 as a check on simulation fidelity, and re-normalization ($\boldsymbol{q} \leftarrow \boldsymbol{q}/\|\boldsymbol{q}\|$ after each integration step) is used as a standard corrective measure.

2.3 Quaternion Kinematics

This section derives the differential equation governing the time evolution of the attitude quaternion, $\dot{\boldsymbol{q}} = f(\boldsymbol{q}, \boldsymbol{\omega})$, which is the kinematic half of the coupled dynamic model integrated in Sec. 5.

2.3.1 Quaternion multiplication

Let two quaternions be written in scalar/vector-part form, $p = (\boldsymbol{p}_v, p_s)$ and $q = (\boldsymbol{q}_v, q_s)$. The quaternion (Hamilton) product $p \otimes q$, which composes the rotation represented by q followed by the rotation represented by p, is

$$p \otimes q = \begin{pmatrix} p_s\boldsymbol{q}_v + q_s\boldsymbol{p}_v + \boldsymbol{p}_v \times \boldsymbol{q}_v \\ p_s q_s - \boldsymbol{p}_v^\mathsf{T}\boldsymbol{q}_v \end{pmatrix}. \tag{2.7}$$

A three-vector $\boldsymbol{v}$ can be embedded in the quaternion algebra as a *pure* quaternion $\bar{\boldsymbol{v}} = (\boldsymbol{v}, 0)$, and the finite-rotation analogue of Eq. (2.1) is $\bar{\boldsymbol{v}}_B = \boldsymbol{q} \otimes \bar{\boldsymbol{v}}_I \otimes \boldsymbol{q}^*$, where $\boldsymbol{q}^* = (-\boldsymbol{\varepsilon}, \eta)$ is the quaternion conjugate (for a unit quaternion, $\boldsymbol{q}^* = \boldsymbol{q}^{-1}$).

2.3.2 Derivation of the kinematic differential equation

Consider the attitude quaternion $\boldsymbol{q}(t)$ that carries vectors from $\mathcal{F}_I$ to $\mathcal{F}_B$, and let $\boldsymbol{\omega}(t)$ be the instantaneous angular velocity of $\mathcal{F}_B$ relative to $\mathcal{F}_I$, expressed in body-frame coordinates. Over an infinitesimal interval dt, the body frame rotates by an infinitesimal angle $\|\boldsymbol{\omega}\|\, dt$ about the instantaneous axis $\boldsymbol{\omega}/\|\boldsymbol{\omega}\|$, as seen *in the body frame itself.* By Eq. (2.4), this infinitesimal rotation is represented by the incremental quaternion

$$\delta \boldsymbol{q} = \begin{pmatrix} \dfrac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|} \sin\left(\dfrac{\|\boldsymbol{\omega}\|\ dt}{2}\right) \\ \cos\left(\dfrac{\|\boldsymbol{\omega}\|\ dt}{2}\right) \end{pmatrix} \xrightarrow[dt \to 0]{} \begin{pmatrix} \frac{1}{2}\boldsymbol{\omega}\, dt \\ 1 \end{pmatrix} + O(dt^2). \tag{2.8}$$

Because $\boldsymbol{\omega}$ is expressed in the (already-rotated) body frame, the incremental rotation is applied by composing it *after* the existing attitude, i.e. by right-multiplication in the sense of Eq. (2.7):

$$\boldsymbol{q}(t + dt) = \boldsymbol{q}(t) \otimes \delta \boldsymbol{q}. \tag{2.9}$$

Physically, Eq. (2.9) states that the orientation at $t + dt$ is obtained by first reaching the orientation at t, then applying the small additional rotation caused by the body's own angular velocity, measured in the frame that is doing the rotating. Substituting Eq. (2.8) into Eq. (2.9), expanding the product with Eq. (2.7) for $p = \boldsymbol{q}(t)$ and $q = \delta \boldsymbol{q}$, keeping only terms to $O(dt)$, and taking the limit $dt \to 0$ gives

$$\dot{\boldsymbol{q}} = \frac{1}{2}\, \boldsymbol{q} \otimes \begin{pmatrix} \boldsymbol{\omega} \\ 0 \end{pmatrix}. \tag{2.10}$$

Expanding the product explicitly using Eq. (2.7) with $p_s = \eta$, $\boldsymbol{p}_v = \boldsymbol{\varepsilon}$, $\boldsymbol{q}_v = \boldsymbol{\omega}$, $q_s = 0$ yields the component form

$$\dot{\boldsymbol{\varepsilon}} = \frac{1}{2}\, (\eta\, \mathbf{I}_3 + \boldsymbol{\varepsilon} mu\ )\boldsymbol{\omega}, \tag{2.11}$$

$$\dot{\eta} = -\frac{1}{2}\, \boldsymbol{\varepsilon}^{\mathsf{T}} \boldsymbol{\omega}. \tag{2.12}$$

Equations (2.11) and (2.12) are combined into the single linear (in $\boldsymbol{\omega}$, for fixed $\boldsymbol{q}$) matrix equation

$$\boxed{\dot{\boldsymbol{q}} = \frac{1}{2}\, \Xi(\boldsymbol{q})\, \boldsymbol{\omega}, \qquad \Xi(\boldsymbol{q}) = \begin{bmatrix} \eta\, \mathbf{I}_3 + \boldsymbol{\varepsilon} mu \\ -\boldsymbol{\varepsilon}^{\mathsf{T}} \end{bmatrix} \in \mathbb{R}^{4\times 3}.} \tag{2.13}$$

Equation (2.13) is the quaternion kinematic differential equation used to propagate attitude throughout this report.

2.3.3 Physical interpretation

The factor of $\frac{1}{2}$ in Eq. (2.13) is a direct consequence of the half-angle relation in Eq. (2.4): the quaternion double-covers the rotation group $SO(3)$ (both $\boldsymbol{q}$ and $-\boldsymbol{q}$ represent the same physical orientation), so a full-rate rotation of the body corresponds to only a half-rate rotation of the quaternion's own representative angle. Equation (2.12) shows that the scalar part decreases fastest when the angular velocity is aligned with the current vector part $\boldsymbol{\varepsilon}$ (i.e., with the Euler rotation axis already in progress), while Eq. (2.11) shows the vector part responding both to $\boldsymbol{\omega}$ directly (through the $\eta\mathbf{I}_3$ term) and to a component of $\boldsymbol{\omega}$ rotated by the current attitude (through the $\boldsymbol{\varepsilon} muterm$).

As a consistency check that also anticipates Sec. 7, differentiating the norm constraint Eq. (2.6) and substituting Eq. (2.13) gives

$$\frac{d}{dt}\left(\boldsymbol{q}^\mathsf{T}\boldsymbol{q}\right) = 2\,\boldsymbol{q}^\mathsf{T}\dot{\boldsymbol{q}} = \boldsymbol{q}^\mathsf{T}\Xi(\boldsymbol{q})\,\boldsymbol{\omega} = (\boldsymbol{\varepsilon} mu\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon} mu\boldsymbol{\varepsilon}\ )^\mathsf{T}\boldsymbol{\omega} = 0, \tag{2.14}$$

using $\boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} mu =$

0^T and cancellation of the $\eta\boldsymbol{\varepsilon}^\mathsf{T}$ terms between the two rows of $\Xi(\boldsymbol{q})$. Equation (2.14) confirms that Eq. (2.13) preserves the unit-norm constraint *exactly* in continuous time for any $\boldsymbol{\omega}(t)$ – the quaternion norm is a conserved quantity of the kinematic equation itself, not an assumption imposed on it. Any drift observed in the numerical simulation of Sec. 6 is therefore attributable purely to the discretization of Eq. (2.13), not to the underlying continuous-time model, which is a useful diagnostic distinction when validating the implementation.

2.4 Rigid Body Rotational Dynamics

While Sec. 2.3 describes how the attitude evolves *given* an angular velocity, it says nothing about what determines that angular velocity. This section derives Euler's rotational equation of motion, which relates $\dot{\boldsymbol{\omega}}$ to the applied torques and the spacecraft's mass distribution.

2.4.1 Angular momentum and the inertia tensor

For a rigid body of mass distribution dm, the angular momentum about the center of mass, expressed in the body frame, is

$$\boldsymbol{H}_B = \int \boldsymbol{r} \times (\boldsymbol{\omega} \times \boldsymbol{r})\,dm = \mathbf{J}\,\boldsymbol{\omega}, \tag{2.15}$$

where $\boldsymbol{r}$ is the position of the mass element relative to the center of mass and

$$\mathbf{J} = \int \left(\left(\boldsymbol{r}^\mathsf{T}\boldsymbol{r}\right)\mathbf{I}_3 - \boldsymbol{r}\boldsymbol{r}^\mathsf{T}\right) dm \tag{2.16}$$

is the inertia tensor: a symmetric, positive-definite 3×3 matrix that is constant in time when expressed in the body frame (because $\mathcal{F}_B$ is rigidly attached to the mass distribution), but generally time-varying if expressed in $\mathcal{F}_I$. Because $\mathbf{J}$ is symmetric, there exists a body-frame orientation – the principal axes – in which $\mathbf{J}$ is diagonal, $\mathbf{J} = \mathrm{diag}(J_x, J_y, J_z)$, with all products of inertia equal to zero [5]. Choosing the body frame $\mathcal{F}_B$ to coincide with the principal axes, as noted in Sec. 2.1, is what allows Eq. (2.16) to be used in its simplest diagonal form in Sec. 3.

2.4.2 Euler's rotational equation of motion

Newton's second law for rotational motion states that the *inertial-frame* time derivative of angular momentum equals the applied external torque, $d\boldsymbol{H}/dt|_{\mathcal{F}_I} = \boldsymbol{\tau}$. Because $\boldsymbol{H}_B$ in Eq. (2.15) is naturally expressed in the rotating body frame, the inertial derivative must be related to the body-frame derivative using the transport theorem for a vector observed from a frame rotating at $\boldsymbol{\omega}$:

$$\left.\frac{d\boldsymbol{H}}{dt}\right|_{\mathcal{F}_I} = \left.\frac{d\boldsymbol{H}}{dt}\right|_{\mathcal{F}_B} + \boldsymbol{\omega} \times \boldsymbol{H}. \tag{2.17}$$

The additional term $\boldsymbol{\omega} \times \boldsymbol{H}$ is not a physical force effect; it is the correction needed because the body-frame basis vectors themselves are rotating, and it has the same mathematical origin as the Coriolis and centripetal terms that appear when translational motion is described in a rotating frame. Substituting Eq. (2.15) into Eq. (2.17) and setting the result equal to $\boldsymbol{\tau}$ gives

$$\boxed{\mathbf{J}\,\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J}\,\boldsymbol{\omega})\,,} \tag{2.18}$$

which is Euler's rotational equation of motion. In component form for the principal-axis inertia tensor $\mathbf{J} = \mathrm{diag}(J_x, J_y, J_z)$, Eq. (2.18) separates into the three familiar scalar equations

$$\begin{aligned} J_x\dot{p} &= \tau_x - (J_z - J_y)\,q\,r, \\ J_y\dot{q} &= \tau_y - (J_x - J_z)\,r\,p, \\ J_z\dot{r} &= \tau_z - (J_y - J_x)\,p\,q. \end{aligned} \tag{2.19}$$

2.4.3 Physical interpretation

The term $\boldsymbol{\omega}\times(\mathbf{J}\boldsymbol{\omega})$ in Eq. (2.18) is usually called the gyroscopic or Euler coupling torque. It vanishes identically only when $\boldsymbol{\omega}$ is aligned with a principal axis or when all three principal moments of inertia are equal. The 3U CubeSat considered here (Sec. 3) is, to a good approximation, axisymmetric about its long (z_B) axis because its cross-section is square ($J_x = J_y \neq J_z$); Eq. (2.19) shows that the spin rate r about the symmetry axis then decouples from p and q under zero applied torque, while p and q remain coupled to one another – the classical torque-free precession of a symmetric rigid body. This coupling is the mathematical origin of the precession and nutation-like behavior expected in rigid body motion, and it is also the reason the attitude controller designed in Sec. 4 must reject a state-dependent nonlinear term, not just track a linear plant.

2.4.4 External torque model

The applied torque in Eq. (2.18) is decomposed as

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\mathrm{ctrl}} + \boldsymbol{\tau}_{\mathrm{dist}}, \tag{2.20}$$

where $\boldsymbol{\tau}_{\mathrm{ctrl}}$ is the commanded reaction wheel torque derived in Sec. 4 and $\boldsymbol{\tau}_{\mathrm{dist}}$ collects environmental disturbance torques (gravity-gradient, residual magnetic dipole, aerodynamic drag, solar radiation pressure). Consistent with the scope defined in **??**, this report sets $\boldsymbol{\tau}_{\mathrm{dist}} = \mathbf{0}$: the simulation in Sec. 5 models only the control torque acting on an otherwise torque-free rigid body. This is a standard first-order simplification for isolating and validating controller behavior, but it is an idealization – in particular, it removes the steady-state disturbance rejection problem that a real CubeSat ADCS must solve continuously – and its consequences are discussed explicitly in Sec. 7.

3 Spacecraft Parameters and Assumptions

Sec. 2 developed the equations of motion in general symbolic form. This section fixes the numerical spacecraft used to exercise those equations in Sec. 5: a 3U CubeSat with a three-axis reaction wheel actuator suite. Every parameter below is either taken directly from the CubeSat Design Specification or computed from an explicitly stated idealized model; none is measured hardware data, and this is stated plainly rather than implied.

3.1 CubeSat Configuration

The spacecraft is a 3-Unit (3U) CubeSat with outer envelope dimensions of

4 Attitude Control System Design

4.1 Control Objective

Given the two-phase scenario defined in **??** – damping an initial post-deployment tumble rate, then slewing to and holding a target inertial attitude – the control task can be stated as a single attitude *regulation* problem: drive the spacecraft to a constant target quaternion $\boldsymbol{q}_d$ with zero angular velocity, and hold it there, starting from an arbitrary initial quaternion $\boldsymbol{q}(0)$ and angular velocity $\boldsymbol{\omega}(0)$. Formally,

$$\lim_{t\to\infty} \boldsymbol{q}(t) = \pm\boldsymbol{q}_d, \qquad \lim_{t\to\infty} \boldsymbol{\omega}(t) = \mathbf{0}, \tag{4.1}$$

where the $\pm$ reflects the double-cover property of the quaternion noted in Sec. 2.2: $\boldsymbol{q}_d$ and $-\boldsymbol{q}_d$ represent the identical physical orientation, so either is an acceptable limit. Because $\boldsymbol{q}_d$ is fixed in $\mathcal{F}_I$ (Sec. 2.1), the desired angular velocity is $\boldsymbol{\omega}_d = \mathbf{0}$ throughout, which is what makes this a regulation problem rather than the more general tracking problem of following a moving target.

4.2 Error Quaternion

Feedback control requires a scalar or vector measure of how far the current attitude is from the target. The attitude error is itself represented as a quaternion, $\boldsymbol{q}_e = (\boldsymbol{\varepsilon}_e, \eta_e)$, defined as

$$\boldsymbol{q}_e = \boldsymbol{q}_d^{-1} \otimes \boldsymbol{q}, \tag{4.2}$$

so that $\boldsymbol{q}_e \to (\mathbf{0}, 1)$ (the identity rotation) exactly when $\boldsymbol{q} \to \pm\boldsymbol{q}_d$. This particular multiplication order – left-multiplying $\boldsymbol{q}$ by the constant $\boldsymbol{q}_d^{-1}$, rather than right-multiplying – is not an arbitrary choice: it is the one that makes $\boldsymbol{q}_e$ obey the same kinematic differential equation as $\boldsymbol{q}$ itself, in terms of the same body angular velocity $\boldsymbol{\omega}$, which is what makes the Lyapunov analysis in Sec. 4.3 possible in closed form. This is shown directly below rather than asserted.

Expanding Eq. (4.2) with the quaternion product rule Eq. (2.7), using $\boldsymbol{q}_d^{-1} = (-\boldsymbol{\varepsilon}_d, \eta_d)$, gives the component form used in the control law below:

$$\boldsymbol{\varepsilon}_e = \eta_d\,\boldsymbol{\varepsilon} - \eta\,\boldsymbol{\varepsilon}_d - \boldsymbol{\varepsilon}_d mu\,\boldsymbol{\varepsilon}, \tag{4.3}$$

$$\eta_e = \eta\eta_d + \boldsymbol{\varepsilon}^{\mathsf{T}}\boldsymbol{\varepsilon}_d. \tag{4.4}$$

As a check, substituting $\boldsymbol{q} = \boldsymbol{q}_d$ into Eqs. (4.3) and (4.4) gives $\boldsymbol{\varepsilon}_e = \mathbf{0}$, $\eta_e = 1$, confirming zero error at the target attitude.

Physically, for a small residual misalignment, $\boldsymbol{\varepsilon}_e \approx \hat{\boldsymbol{e}}_e\,(\phi_e/2)$ by Eq. (2.4), so the vector part of the error quaternion is approximately half the (small) rotation vector still needed to reach the target – this is what makes $\boldsymbol{\varepsilon}_e$ a natural proportional error signal for the control law in Sec. 4.3. The scalar part $\eta_e = \cos(\phi_e/2)$ indicates how large the remaining rotation angle ϕ_e is: η_e near $+1$ means a small residual rotation, while η_e near -1 means the remaining rotation is close to a full 360° – a case addressed explicitly in Sec. 4.3.

Equation (4.2) obeys a clean kinematic equation because $\boldsymbol{q}_d$ is constant and quaternion multiplication is associative. Differentiating Eq. (4.2) and substituting the kinematic equation Eq. (2.10) for $\dot{\boldsymbol{q}}$,

$$\dot{\boldsymbol{q}}_e = \boldsymbol{q}_d^{-1} \otimes \dot{\boldsymbol{q}} = \boldsymbol{q}_d^{-1} \otimes \left(\tfrac{1}{2}\boldsymbol{q} \otimes \begin{pmatrix}\boldsymbol{\omega}\\ 0\end{pmatrix}\right) = \tfrac{1}{2}\left(\boldsymbol{q}_d^{-1} \otimes \boldsymbol{q}\right) \otimes \begin{pmatrix}\boldsymbol{\omega}\\ 0\end{pmatrix} = \tfrac{1}{2}\,\boldsymbol{q}_e \otimes \begin{pmatrix}\boldsymbol{\omega}\\ 0\end{pmatrix}, \tag{4.5}$$

where the middle step only regroups the product $(\boldsymbol{q}_d^{-1} \otimes \boldsymbol{q} \otimes (\boldsymbol{\omega}, 0))$ by associativity – it does not require $\boldsymbol{q}_d$ to be constant to be valid algebra, but constancy of $\boldsymbol{q}_d$ is what makes $\dot{\boldsymbol{q}}_d^{-1}$ vanish in the first step. Equation (4.5) shows $\boldsymbol{q}_e$ satisfies exactly the same kinematic form as $\boldsymbol{q}$, *with the same* $\boldsymbol{\omega}$:

$$\dot{\boldsymbol{q}}_e = \tfrac{1}{2}\Xi(\boldsymbol{q}_e)\,\boldsymbol{\omega}, \tag{4.6}$$

which is used directly in the stability analysis of Sec. 4.3. This clean result is a consequence of the regulation scenario ($\boldsymbol{q}_d$ constant, i.e. $\boldsymbol{\omega}_d = \mathbf{0}$, Sec. 4.1); for a moving target ($\boldsymbol{\omega}_d \neq \mathbf{0}$) an additional term involving $\boldsymbol{\omega}_d$ would appear.

4.3 PD Controller Formulation

4.3.1 Control law

A proportional-derivative control law is formed directly on the quaternion vector error $\boldsymbol{\varepsilon}_e$ and the body angular velocity $\boldsymbol{\omega}$ (equivalently the angular velocity error, since $\boldsymbol{\omega}_d = \mathbf{0}$):

$$\boxed{\boldsymbol{\tau}_{\text{ctrl}} = -K_p\,\text{sgn}(\eta_e)\,\boldsymbol{\varepsilon}_e - K_d\,\boldsymbol{\omega},} \tag{4.7}$$

with scalar gains $K_p, K_d > 0$. The proportional term drives the vector error $\boldsymbol{\varepsilon}_e$ toward zero (i.e., drives the attitude toward $\boldsymbol{q}_d$), while the derivative term damps the angular velocity, both consistent with the regulation objective Eq. (4.1). The $\operatorname{sgn}(\eta_e)$ factor (defined as $+1$ if $\eta_e \geq 0$ and -1 otherwise) implements the standard *shortest-path* correction: because $\boldsymbol{q}_e$ and $-\boldsymbol{q}_e$ represent the same physical error, and Eq. (4.3) shows that $\boldsymbol{\varepsilon}_e$ changes sign along with the overall sign of $\boldsymbol{q}_e$, applying this sign correction ensures the controller always commands the rotation corresponding to the error angle $\phi_e \in (-180°, 180°]$, rather than potentially commanding a needlessly long rotation when $\eta_e < 0$. This is not a cosmetic detail: without it, the closed-loop system exhibits the well-documented *unwinding phenomenon*, in which a spacecraft starting arbitrarily close to the target attitude (but on the $\eta_e < 0$ branch) is driven through a nearly complete, physically unnecessary revolution before converging [6, 7].

4.3.2 Lyapunov stability analysis

Stability of Eq. (4.7) can be shown directly from Euler's equation Eq. (2.18) and the error quaternion kinematics, without linearizing the dynamics. Consider the candidate Lyapunov function

$$V(\boldsymbol{\varepsilon}_e, \eta_e, \boldsymbol{\omega}) = 2K_p\,(1 - |\eta_e|) + \frac{1}{2}\boldsymbol{\omega}^\mathsf{T}\mathbf{J}\,\boldsymbol{\omega}, \tag{4.8}$$

which is positive definite about the equilibrium $(\boldsymbol{\varepsilon}_e, \boldsymbol{\omega}) = (\mathbf{0}, \mathbf{0})$: the first term is zero only when $|\eta_e| = 1$ (i.e., $\boldsymbol{\varepsilon}_e = \mathbf{0}$, since $\boldsymbol{\varepsilon}_e^\mathsf{T}\boldsymbol{\varepsilon}_e + \eta_e^2 = 1$), and the second term is zero only when $\boldsymbol{\omega} = \mathbf{0}$, since $\mathbf{J}$ is positive definite (Sec. 2.4). Differentiating, using $\dot{\eta}_e = -\frac{1}{2}\boldsymbol{\varepsilon}_e^\mathsf{T}\boldsymbol{\omega}$ (the scalar-part row of Eq. (4.6), derived in Sec. 4.2) together with $\operatorname{sgn}(\eta_e)\dot{\eta}_e = \frac{d}{dt}|\eta_e|$, and using Euler's equation Eq. (2.18) for $\mathbf{J}\dot{\boldsymbol{\omega}}$:

$$\begin{aligned}\dot{V} &= -2K_p \operatorname{sgn}(\eta_e)\dot{\eta}_e + \boldsymbol{\omega}^\mathsf{T}\mathbf{J}\dot{\boldsymbol{\omega}} \\ &= K_p \operatorname{sgn}(\eta_e)\,\boldsymbol{\varepsilon}_e^\mathsf{T}\boldsymbol{\omega} + \boldsymbol{\omega}^\mathsf{T}\left(\boldsymbol{\tau}_{\text{ctrl}} - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})\right).\end{aligned} \tag{4.9}$$

The gyroscopic term drops out identically because $\boldsymbol{\omega}^\mathsf{T}(\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega}) = 0$ for any vector triple product of this form. Substituting the control law Eq. (4.7) for $\boldsymbol{\tau}_{\text{ctrl}}$:

$$\dot{V} = K_p \operatorname{sgn}(\eta_e)\,\boldsymbol{\varepsilon}_e^\mathsf{T}\boldsymbol{\omega} - K_p \operatorname{sgn}(\eta_e)\,\boldsymbol{\omega}^\mathsf{T}\boldsymbol{\varepsilon}_e - K_d\,\boldsymbol{\omega}^\mathsf{T}\boldsymbol{\omega} = -K_d\,\|\boldsymbol{\omega}\|^2 \leq 0, \tag{4.10}$$

since the two proportional-term contributions are equal scalars and cancel exactly, independent of the specific values of $\boldsymbol{\varepsilon}_e$ or $\mathbf{J}$. Equation (4.10) shows the closed-loop system is Lyapunov stable for any $K_p, K_d > 0$. Because $\dot{V} \equiv 0$ only on the set $\boldsymbol{\omega} = \mathbf{0}$, and Eq. (2.18) shows that $\boldsymbol{\omega} \equiv \mathbf{0}$ can only be an invariant trajectory if $\boldsymbol{\tau}_{\text{ctrl}} \equiv \mathbf{0}$, which by Eq. (4.7) requires $\boldsymbol{\varepsilon}_e \equiv \mathbf{0}$, LaSalle's invariance principle gives asymptotic convergence to $\boldsymbol{\varepsilon}_e = \mathbf{0}, \boldsymbol{\omega} = \mathbf{0}$ [7, 2]. This confirms, without approximation, that the nonlinear gyroscopic coupling identified in Sec. 2.4 does not destabilize the PD law – it is annihilated exactly in $\dot{V}$ by the same triple-product identity that made it absent from the rotational kinetic energy rate in the first place.

It is worth being precise about what this result does and does not establish. It shows *local* asymptotic stability of the target attitude is in fact *almost-global*: the only other candidate equilibrium admitted by the argument above is $\boldsymbol{\varepsilon}_e = \mathbf{0}$ with $\eta_e = -\mathrm{sgn}(\eta_e(0))$, i.e., the antipodal representation of the identity rotation, which corresponds to the same physical attitude, not a genuine second target. The $\mathrm{sgn}(\eta_e)$ term in Eq. (4.7) exists precisely to keep the system on the correct branch and avoid transiently approaching that antipodal point. The topological result of Bhat and Bernstein [6] shows that no *continuous* state feedback law can achieve a truly global, unwinding-free equilibrium on $SO(3)$; the sign-switched law Eq. (4.7) is the standard, discontinuous (at $\eta_e = 0$) way around that obstruction, and is why the switching term is included rather than treated as optional.

4.4 Control Torque Generation

Equation (4.7) produces a commanded torque vector $\boldsymbol{\tau}_{\text{ctrl}} \in \mathbb{R}^3$ expressed in the body principal-axis frame. Because the actuator suite defined in Sec. 3.1 consists of exactly three reaction wheels, one aligned with each body principal axis, the mapping from the commanded torque vector to individual wheel motor torque commands is the identity (up to the reaction-torque sign convention): each wheel is commanded to produce a motor torque equal and opposite to the corresponding component of $\boldsymbol{\tau}_{\text{ctrl}}$, so that the reaction torque delivered to the spacecraft body equals $\boldsymbol{\tau}_{\text{ctrl}}$ component-by-component. No control allocation (pseudo-inverse mixing matrix) is required, which is a direct consequence of choosing a minimal, non-redundant, axis-aligned wheel configuration in Sec. 3.1; a four-wheel pyramid configuration, commonly used operationally for single-failure redundancy, would instead require solving an underdetermined allocation problem, which is noted as a possible extension in Sec. 7.

Consistent with the actuator assumptions of **??**, each component of $\boldsymbol{\tau}_{\text{ctrl}}$ is saturated independently at the wheel's maximum torque before being applied in the simulation:

$$\tau_{\text{applied},i} = \max\left(-\tau_{\max},\ \min\left(\tau_{\max},\ \tau_{\text{ctrl},i}\right)\right), \qquad i \in \{x, y, z\}, \tag{4.11}$$

with $\tau_{\max}$ given in **??**. Saturation is the one actuator nonideality retained in this study (**??**); it is included because it directly bounds the achievable initial deceleration of the detumble phase in Sec. 5 and is therefore necessary for the simulated scenario to be self-consistent, even though the underlying wheel momentum-storage dynamics are not modeled.

5 Simulation Methodology

5.1 MATLAB Implementation Architecture

The closed-loop model of Secs. 2 and 4 is implemented in MATLAB as a small set of single-purpose functions rather than one monolithic script, so that each block corresponds directly to one derived equation and can be unit-checked independently. Table 1 summarizes the intended module breakdown.

Table 1: MATLAB implementation modules and their corresponding equations.

File	Purpose
`cubesat_params.m`	Defines mass properties, inertia tensor, and wheel limits (**????**).
`quat_utils.m`	Quaternion product, conjugate, and quaternion-to-DCM utilities (Eq. (2.7) and **??**).
`attitude_dynamics.m`	State derivative function combining quaternion kinematics (Eq. (2.13)) and Euler's equation (Eq. (2.18)) into a single 7-state ODE right-hand side.
`pd_controller.m`	Computes the error quaternion (Eqs. (4.3) and (4.4)), evaluates the PD law (Eq. (4.7)), and applies torque saturation (Eq. (4.11)).
`rk4_integrator.m`	Fixed-step RK4 propagation of the state, with quaternion renormalization (Eq. (2.6)) applied once per step.
`main_simulation.m`	Driver script: sets initial conditions and gains, runs the integration loop, and calls the plotting routines used in Sec. 6.

The spacecraft state is collected into a single 7-element vector $\boldsymbol{x} = [\boldsymbol{q}^{\mathsf{T}},\ \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^7$, and `attitude_dynamics.m` implements the combined right-hand side

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{\tau}_{\text{ctrl}}) = \begin{bmatrix} \frac{1}{2}\Xi(\boldsymbol{q})\,\boldsymbol{\omega} \\ \mathbf{J}^{-1}\left(\boldsymbol{\tau}_{\text{ctrl}} - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})\right) \end{bmatrix}, \tag{5.1}$$

directly combining Eqs. (2.13) and (2.18). Because $\boldsymbol{\tau}_{\text{ctrl}}$ in Eq. (5.1) is itself a function of the current state through Eq. (4.7), the controller is evaluated once per integration stage (i.e., `pd_controller.m` is called from inside `attitude_dynamics.m` or immediately before it) so that the closed-loop system, not an open-loop torque profile, is what is actually integrated.

5.2 Numerical Integration Method

The state equation Eq. (5.1) is propagated using the classical fourth-order Runge-Kutta method (RK4) with a fixed time step Δt. RK4 is chosen over a variable-step solver (e.g., MATLAB's `ode45`) as the primary integrator for three reasons specific to this problem: (i) the dynamics in Eq. (5.1) are smooth and non-stiff – there are no fast unmodeled structural or actuator modes (**??**) that would otherwise force a variable-step solver down to very small steps – so a fixed step is not a computational penalty here; (ii) a fixed, known step size makes it straightforward to apply the quaternion re-normalization of Eq. (2.6) at a consistent, controlled cadence and to reason about its effect on norm drift, which is one of the validation checks in Sec. 7; and (iii) fixed-step results are trivially reproducible without depending on a particular solver's internal step-size heuristics, which matters for a report whose figures must be regenerable. Algorithm 1 summarizes the per-step update.

Algorithm 1 Closed-loop RK4 integration step

function RK4Step($\boldsymbol{x}_k$, Δt)
  $\boldsymbol{k}_1 \leftarrow f(\boldsymbol{x}_k)$ ▷ f internally evaluates Eq. (4.7) at the given state
  $\boldsymbol{k}_2 \leftarrow f(\boldsymbol{x}_k + \frac{\Delta t}{2}\boldsymbol{k}_1)$
  $\boldsymbol{k}_3 \leftarrow f(\boldsymbol{x}_k + \frac{\Delta t}{2}\boldsymbol{k}_2)$
  $\boldsymbol{k}_4 \leftarrow f(\boldsymbol{x}_k + \Delta t\, \boldsymbol{k}_3)$
  $\boldsymbol{x}_{k+1} \leftarrow \boldsymbol{x}_k + \frac{\Delta t}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right)$
  $\boldsymbol{q}_{k+1} \leftarrow \boldsymbol{q}_{k+1} / \left\|\boldsymbol{q}_{k+1}\right\|$ ▷ re-normalize per Eq. (2.6)
  return $\boldsymbol{x}_{k+1}$
end function

As a cross-check on the fixed-step RK4 results, Sec. 7 additionally compares against MATLAB's adaptive `ode45` solver applied to the same right-hand side Eq. (5.1); close agreement between the two independent integration schemes is used as evidence that the reported transient behavior in Sec. 6 reflects the underlying dynamics rather than an artifact of the chosen integrator or step size.

5.3 Controller Gain Selection

Sec. 4.3 established stability of Eq. (4.7) for *any* $K_p, K_d > 0$, but stability alone does not fix a specific response speed. Gains are selected here from an approximate linearization valid for small attitude error. For $\eta_e \approx 1$ and $\boldsymbol{\varepsilon}_e$ small, Eq. (2.11) reduces to $\dot{\boldsymbol{\varepsilon}}_e \approx \frac{1}{2}\boldsymbol{\omega}$, and Eq. (2.18) reduces to $\mathbf{J}\dot{\boldsymbol{\omega}} \approx \boldsymbol{\tau}_{\text{ctrl}}$ (the gyroscopic term is second-order small near equilibrium). Substituting the control law Eq. (4.7) (with $\operatorname{sgn}(\eta_e) = 1$ near equilibrium) and eliminating $\boldsymbol{\omega}$ gives, per axis with scalar inertia J,

$$\ddot{q}_e + \frac{K_d}{J}\dot{q}_e + \frac{K_p}{2J}\, q_e = 0, \tag{5.2}$$

a standard second-order form with natural frequency $\omega_n = \sqrt{K_p/2J}$ and damping ratio $\zeta = K_d/(2\sqrt{2K_pJ})$. Solving for the gains that place ω_n, ζ at chosen design values,

$$K_p = 2J\omega_n^2, \qquad K_d = 2\zeta\omega_n J. \tag{5.3}$$

Targeting a critically-damped-like response ($\zeta = 0.7$) with a 2%-settling time of approximately $t_s \approx 4/(\zeta\omega_n) = 60\,\text{s}$ – a conservative, low-torque response appropriate for the small actuator budget in **??** – gives $\omega_n \approx 0.095\,\text{rad/s}$. Using the larger, transverse principal inertia $J = J_x = J_y = 0.0333\,\text{kg}\,\text{m}^2$ from **??** in Eq. (5.3) (the more torque-demanding axis pair) yields the gains in Table 2, applied identically to all three axes per the scalar-gain law Eq. (4.7).

With these gains, the proportional term alone never exceeds $K_p \cdot 1 = 6.0 \times 10^{-4}\,\text{N}\,\text{m}$ (since $\|\boldsymbol{\varepsilon}_e\| \leq 1$), leaving margin below $\tau_{\max} = 1.0 \times 10^{-3}\,\text{N}\,\text{m}$ (**??**) for the derivative term to act on the initial tumble rate defined in Sec. 5.4 – Sec. 6 shows this initial-rate torque does briefly saturate at $\tau_{\max}$, which is retained deliberately (rather than tuned away) so that the saturation logic of Eq. (4.11) is actually exercised and its effect on convergence can be discussed in Sec. 7.

Table 2: Adopted PD controller gains.

Parameter	Symbol	Value
Proportional gain	K_p	6.0×10^{-4} N m
Derivative gain	K_d	4.4×10^{-3} N m s

5.4 Simulation Scenario

The scenario combines the post-deployment detumble and slew-to-target phases identified in **??** into a single continuous maneuver rather than two sequential control modes. This is a direct consequence of the actuator choice: a reaction-wheel PD regulator (Eq. (4.7)) acts on $\boldsymbol{\varepsilon}_e$ and $\boldsymbol{\omega}$ simultaneously, so it damps an initial tumble rate and drives the attitude error toward zero in the same continuous control action. This differs from a magnetorquer-only architecture, where detumbling is typically handled by a distinct rate-only law (e.g., B-dot) before a separate pointing controller is engaged, because magnetic actuation cannot always produce an arbitrary commanded torque direction at a given instant – the instantaneous control torque is constrained to lie in the plane perpendicular to the local geomagnetic field vector [8] – so Sec. 4.4 does not apply to that actuator type. With three-axis reaction wheels, that architectural split is unnecessary, and the "two-phase" scenario is realized here as a single transient with an initially dominant rate-damping response that smoothly gives way to a dominant attitude-error response as $\boldsymbol{\omega} \to \mathbf{0}$.

5.5 Initial Conditions

Table 3 lists the adopted initial state and target attitude. The spacecraft is initialized coincident with $\mathcal{F}_I$ ($\boldsymbol{q}(0)$ equal to the identity rotation) and given a representative post-deployment tumble rate; the target attitude $\boldsymbol{q}_d$ corresponds to a single 120° rotation about the body-diagonal axis $\hat{\boldsymbol{e}} = \frac{1}{\sqrt{3}}[1,1,1]^\mathsf{T}$, chosen specifically as a large-angle, non-coordinate-axis maneuver of the kind that is well-behaved for a quaternion representation but would pass close to a coordinate singularity for at least one possible Euler angle sequence – directly exercising the representational advantage argued for in **??**.

Table 3: Simulation initial conditions and target attitude.

Quantity	Symbol	Value
Initial quaternion	$\boldsymbol{q}(0)$	$[0,\ 0,\ 0,\ 1]^\mathsf{T}$ (identity)
Initial angular velocity	$\boldsymbol{\omega}(0)$	$[10,\ -15,\ 8]^\mathsf{T}$ deg/s ($[0.175,\ -0.262,\ 0.140]^\mathsf{T}$ rad/s)
Target quaternion	$\boldsymbol{q}_d$	$[0.5,\ 0.5,\ 0.5,\ 0.5]^\mathsf{T}$ (120° about $\frac{1}{\sqrt{3}}[1,1,1]^\mathsf{T}$)
Simulation duration	T	120 s
Integration step size	Δt	0.01 s

The step size $\Delta t = 0.01\,\mathrm{s}$ is roughly two orders of magnitude smaller than both the control bandwidth timescale ($1/\omega_n \approx 10.5\,\mathrm{s}$, Sec. 5.3) and the initial tumble rotation timescale ($\sim 1/\|\boldsymbol{\omega}(0)\| \approx 2.9\,\mathrm{s}$), which is a standard oversampling margin for a fixed-step RK4 integration of a non-stiff system of this bandwidth and is refined if needed once the `ode45` cross-check of Sec. 5.2 is available.

6 Results and Analysis

This section presents the output of the simulation defined in Sec. 5: the 3U CubeSat, initialized at the identity attitude with a $[10, -15, 8]^{\mathsf{T}}$ deg/s tumble (Table 3), is driven by the PD controller of Eq. (4.7) toward the 120° target attitude $\boldsymbol{q}_d = [0.5, 0.5, 0.5, 0.5]^{\mathsf{T}}$. Table 4 collects the key numerical outcomes referenced throughout this section; all values are taken directly from the RK4 simulation output.

Table 4: Key simulation outcomes.

Metric	Value
Settling time ($\|\boldsymbol{\varepsilon}_e\| < 0.02$ and $\|\boldsymbol{\omega}\| < 0.01\,\mathrm{rad/s}$, sustained)	$74.1\,\mathrm{s}$
Final attitude error, $\|\boldsymbol{\varepsilon}_e\|$	9.6×10^{-4} (0.11° equiv.)
Final angular velocity, $\|\boldsymbol{\omega}\|$	1.8×10^{-4} rad/s (0.0103°/s)
Peak applied torque	$1.0\,\mathrm{mN\,m}$ ($= \tau_{\max}$)
Fraction of run with any axis saturated	4.7% (first $\approx 5.6\,\mathrm{s}$)
Max. quaternion norm error, $\lvert\|\boldsymbol{q}\| - 1\rvert$	2.2×10^{-16} (machine precision)
Max. RK4/`ode45` attitude discrepancy	$1.3 \times 10^{-4}\,{}^{\circ}$

6.1 Quaternion Response

Fig. 1 shows all four quaternion components converging from the initial identity attitude toward the target $\boldsymbol{q}_d = [0.5, 0.5, 0.5, 0.5]^{\mathsf{T}}$ (dashed lines). The response is not a simple monotonic first-order approach: q_2 initially overshoots to approximately -0.58 before turning around and climbing to its target value of 0.5, and q_1, q_3 show comparable, though smaller, non-monotonic excursions before all four components settle within the final few percent of the simulation. This behavior is consistent with the coupled, nonlinear character of Eq. (2.18) discussed in Sec. 2.4: because the maneuver combines a large initial angular rate with a large attitude error simultaneously, the transient is not well described by the small-angle, single-axis linearization used only to *select* the gains in Sec. 5.3 – that linearization was never claimed to predict the large-angle transient shape, only to size the gains, and Fig. 1 shows the resulting closed-loop response is qualitatively more complex than the linear design model, while still converging to the correct target.

Fig. 2 shows the same trajectory re-expressed as equivalent 3-2-1 Euler angles via **??** and the standard DCM-to-Euler-angle inversion, purely as a diagnostic and a physical-interpretation aid

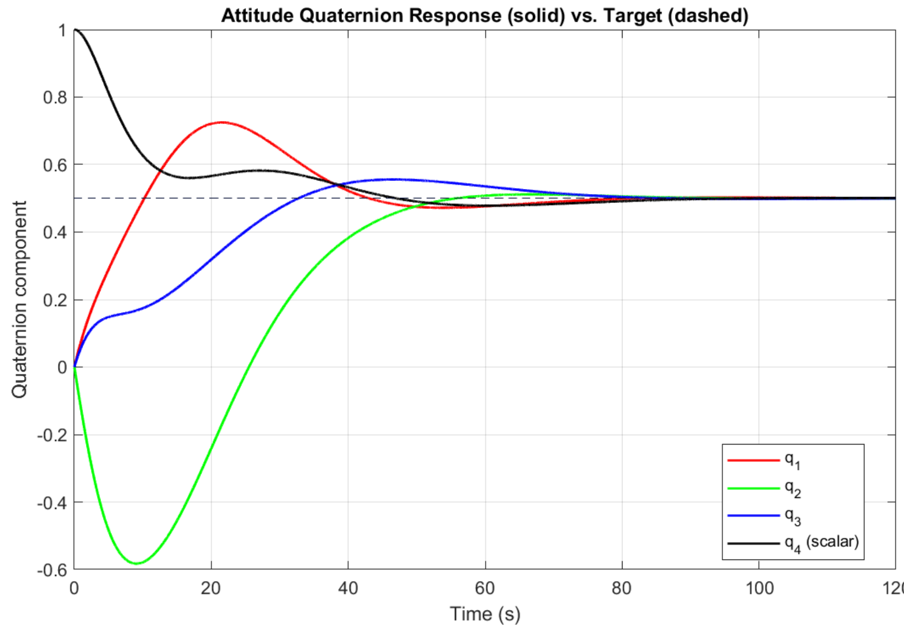

Figure 1: Attitude quaternion components (solid) versus target values (dashed).

– the simulation itself never uses Euler angles internally (**??**). The roll angle ϕ overshoots to approximately 118° before settling near 90°, and the pitch angle θ swings to about $-68°$ before returning to 0°; the final orientation $(\phi, \theta, \psi) = (90°, 0°, 90°)$ is consistent with the target DCM computed directly from $\boldsymbol{q}_d$ in Sec. 6.1 being the coordinate-permutation matrix $\left[\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right]$, itself a direct consequence of $\boldsymbol{q}_d$ representing exactly 120° about the body diagonal $\frac{1}{\sqrt{3}}[1, 1, 1]^\mathsf{T}$ (a rotation that cyclically permutes the coordinate axes). For this particular trajectory, θ does not reach the $\pm 90°$ singularity identified in Sec. 2.2; nonetheless, the Euler angle history exhibits substantially larger and non-monotonic excursions (118° roll overshoot, 68° pitch swing) than the quaternion components in Fig. 1, which remain bounded in $[-1, 1]$ throughout by construction – illustrating, empirically rather than as an assumed worst case, why a representation immune to coordinate singularities was preferred for this class of maneuver.

6.2 Angular Velocity Response

Fig. 3 shows the body angular velocity decaying from the initial tumble $[10, -15, 8]^\mathsf{T}$ deg/s to within 0.01°/s of zero by the end of the simulation. As with the quaternion response, the decay is not a clean single-time-constant exponential: ω_x shows a distinct shoulder near $t \approx$ 5-15 s where its decay rate slows before resuming, and ω_y actually reverses sign and grows to a secondary local peak of about 4.5 deg/s near $t \approx 23$ s before finally decaying. This is the direct dynamical signature of the gyroscopic coupling term $\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})$ in Eq. (2.18): because the inertia tensor is axisymmetric but not isotropic (**??**), angular momentum is exchanged between the transverse axes during the transient exactly as anticipated qualitatively in Sec. 2.4.

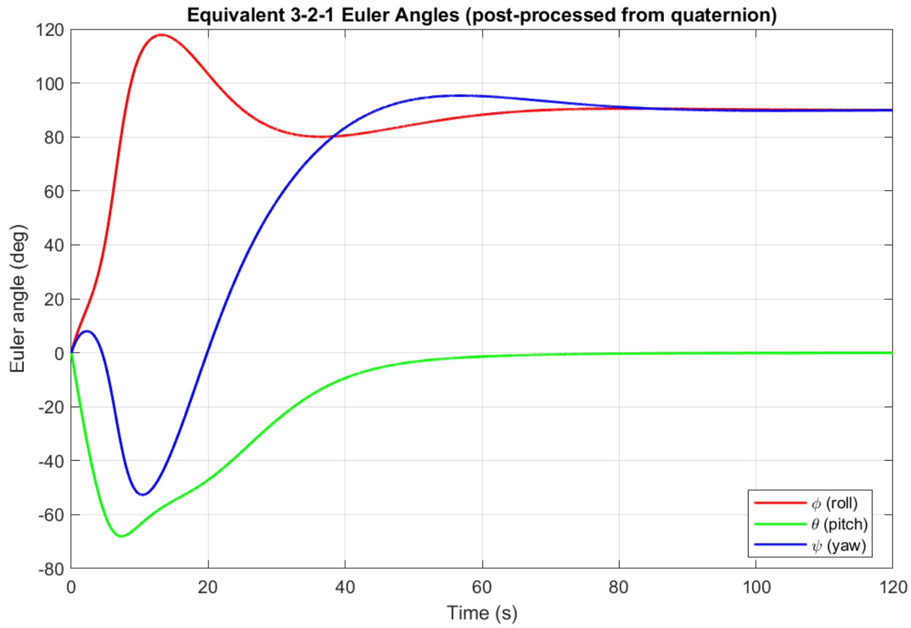

Figure 2: Equivalent 3-2-1 Euler angles, post-processed from the quaternion trajectory for interpretation only.

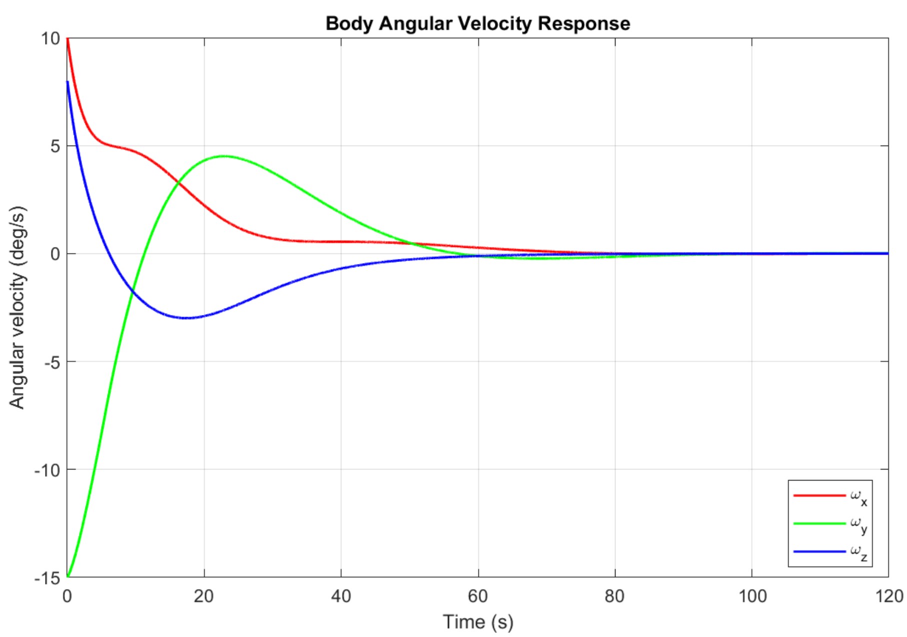

Figure 3: Body angular velocity response.

6.3 Control Torque Behavior

Fig. 4 shows the applied (saturated) reaction wheel torque on all three axes. The y-axis wheel, which must counter the largest initial rate component ($\omega_y(0) = -15\,\mathrm{deg/s}$), saturates at $\tau_{\max} = 1.0\,\mathrm{mN\,m}$ for approximately the first 5.6 s of the simulation (4.7% of the total run, Table 4), confirming the deliberate design choice flagged in Sec. 5.3: the initial-rate torque demand does exceed the wheel's authority briefly, and the saturation logic of Eq. (4.11) is genuinely exercised rather than being a dormant code path. After this initial interval, all three torque components settle into smooth, unsaturated transients before decaying to zero, consistent with the sub-saturation torque margin computed in Sec. 5.3 for the proportional term alone.

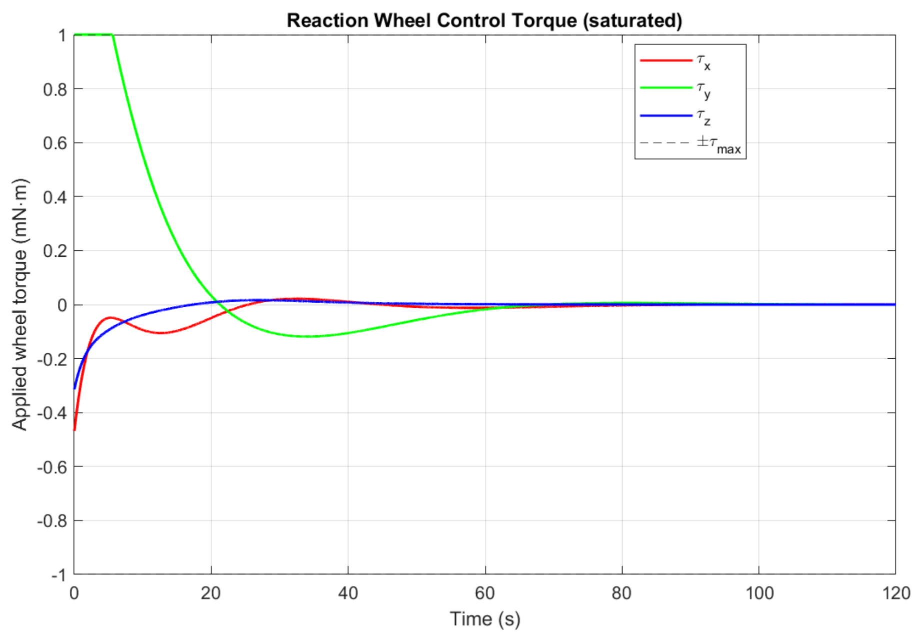

Figure 4: Applied reaction wheel torque per axis, with saturation limits shown as dashed lines.

6.4 Stability Analysis

Fig. 5 plots the Lyapunov function $V(t)$ from Eq. (4.8) evaluated along the simulated trajectory, on a logarithmic scale. $V(t)$ is monotonically non-increasing over the entire 120 s run, decreasing by roughly six orders of magnitude, with a visible change in slope near $t \approx$ 45-65 s corresponding to the interval where $\|\boldsymbol{\omega}\|$ passes through its own local dip in Fig. 3 (recall $\dot{V} = -K_d \|\boldsymbol{\omega}\|^2$ exactly, Eq. (4.10), so $\dot{V}$ is small precisely when $\|\boldsymbol{\omega}\|$ is small, without V ever increasing). This is a direct, non-trivial confirmation of the analytical result derived in Sec. 4.3: the closed-loop simulation obeys the exact stability certificate produced by the nonlinear Lyapunov analysis, not merely the qualitative expectation that the controller should eventually converge.

This result was not obtained on the first implementation attempt. An earlier version of the controller code defined the error quaternion using the opposite quaternion multiplication order,

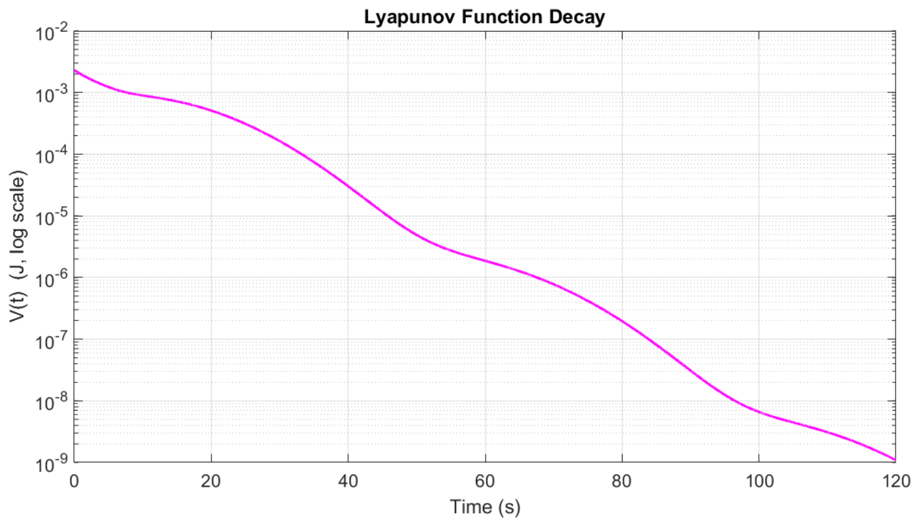

Figure 5: Lyapunov function $V(t)$ (log scale), confirming monotonic decrease predicted by Eq. (4.10).

$\boldsymbol{q}_e = \boldsymbol{q} \otimes \boldsymbol{q}_d^{-1}$ rather than Eq. (4.2); that version is algebraically valid as a measure of attitude error, but it does *not* satisfy Eq. (4.6) for the same $\boldsymbol{\omega}$, and the resulting $V(t)$ computed from that simulation was non-monotonic, alternately decreasing and increasing over multiple 20-40 s intervals despite Eq. (4.10) guaranteeing $\dot{V} \leq 0$ unconditionally. This mismatch between a theoretical guarantee and simulated behavior was the signal that the implementation, not the guarantee, was wrong, and tracing it back through the derivation in Sec. 4.2 located the multiplication-order error described there. This episode is included here deliberately: a Lyapunov function that fails to decrease monotonically in simulation is one of the most direct available checks that a nonlinear attitude controller implementation matches its own derivation, and it is a check worth running on any such implementation before trusting its transient results.

7 Validation and Discussion

7.1 Comparison Against Expected Physical Behavior

Three independent checks were used to test whether the simulation in Sec. 6 reflects the model actually derived in Secs. 2 and 4, rather than an artifact of the implementation.

Quaternion norm conservation. Sec. 2.3 proved analytically that the continuous-time kinematic equation Eq. (2.13) conserves $\|\boldsymbol{q}\| = 1$ exactly (Eq. (2.14)), and attributed any drift in simulation to discretization rather than the underlying model. Fig. 6 shows $|\|\boldsymbol{q}\| - 1|$ over the full run: it remains at the level of 10^{-16}-10^{-18} throughout, i.e., double-precision floating-point round-off, with no visible growth trend over 120 s and 12,000 integration steps. This is the expected result given the per-step renormalization in Algorithm 1: it does not test whether Eq. (2.13) itself is correctly implemented (an error there could still preserve the norm while integrating the wrong direction), but it does confirm that the renormalization strategy adopted specifically to bound norm drift (Sec. 2.2) works as intended.

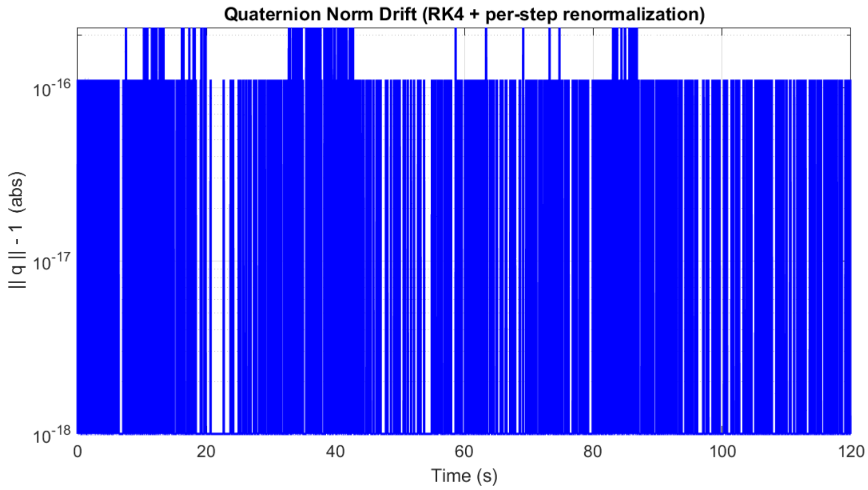

Figure 6: Quaternion norm error over the simulation, remaining at floating-point round-off level.

Independent integrator cross-check. Sec. 5.2 proposed comparing the primary fixed-step RK4 solution against MATLAB's adaptive `ode45` solver applied to the identical right-hand side Eq. (5.1), as a check that the reported transient is not an artifact of the chosen step size or integration scheme. Fig. 7 shows the attitude discrepancy between the two solutions, computed from the quaternion dot product, over the full run: the maximum discrepancy is 1.3×10^{-4} degrees, five to six orders of magnitude smaller than any feature of interest in Sec. 6 (e.g., the $\sim 120°$ overall maneuver or the $0.11°$ final error). This is strong evidence that the qualitative transient behavior discussed in Secs. 6.1 and 6.2 – the overshoot, the secondary ω_y peak – reflects genuine closed-loop dynamics rather than numerical error, since two structurally different integrators with independent error mechanisms agree to this precision.

Lyapunov monotonicity. As discussed already in Sec. 6.4, the simulated $V(t)$ decreases monotonically, matching Eq. (4.10) exactly. This is arguably the strongest of the three checks, because it tests the *closed-loop coupling* between the controller and the plant directly (both Eqs. (2.18) and (4.7) have to be implemented consistently with each other for the exact cancellation in Eq. (4.10) to hold), and, as recounted in Sec. 6.4, it is what actually caught a genuine implementation defect (an error-quaternion multiplication-order bug) during the development of this simulation – a concrete illustration of why a derived stability certificate is worth checking numerically, not just citing.

Settling time versus the linear design prediction. Sec. 5.3 targeted a settling time of approximately 60 s from a small-angle linearization that explicitly neglected the gyroscopic term. The simulated settling time is 74.1 s (Table 4), about 23% longer. This gap is consistent with, and

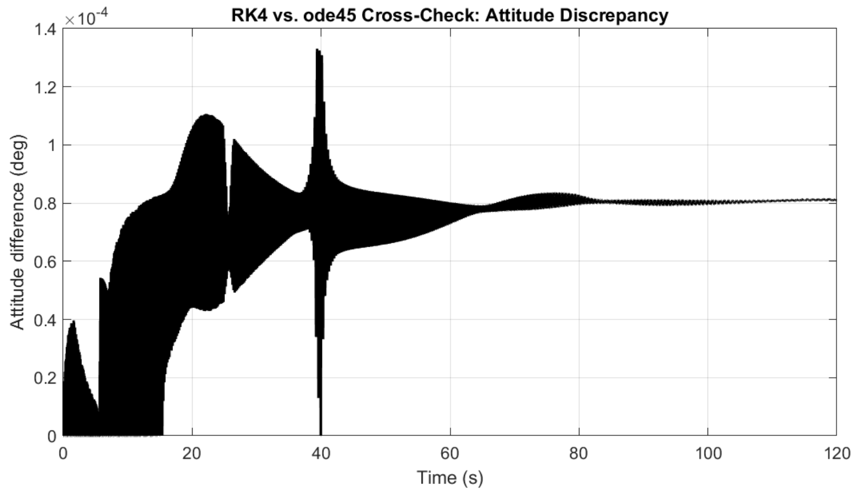

Figure 7: Attitude discrepancy between the primary fixed-step RK4 solution and an independent `ode45` solution of the same right-hand side.

expected from, the limitations of that linearization stated at the time it was derived: the actual maneuver combines a 120° initial attitude error with a $\sim$19.7°/s initial tumble simultaneously, well outside the small-$\boldsymbol{\varepsilon}_e$, small-$\boldsymbol{\omega}$ regime the linear formula Eq. (5.2) assumes, and the wheel saturates for the first 5.6 s (Sec. 6.3), further slowing the response relative to the unsaturated linear prediction. The direction and rough magnitude of the discrepancy (slower than predicted, not faster or unstable) are physically sensible: both the neglected gyroscopic coupling and the saturation act to remove usable control authority from the linear design's assumptions, not add extra authority.

7.2 Limitations

?? ranked the modeling assumptions of Sec. 3 by expected impact before any simulation existed; the results in Sec. 6 make it possible to revisit that ranking with evidence rather than only judgment, and to add limitations specific to the control design and validation scope actually exercised.

- **No disturbance torques (confirmed highest-impact).** The controller in this report has demonstrated convergence to a fixed target in a torque-free environment. It has not demonstrated *disturbance rejection* – holding a target attitude against a persistent, non-zero external torque – which is the steady-state operating condition of a real CubeSat and was already flagged in **??** as the assumption most in need of relaxation.
- **Idealized reaction wheel momentum storage.** The wheel saturation event in Sec. 6.3 shows that even the limited actuator nonideality retained in this study (a static torque limit) already materially affects the transient (Sec. 7.1). A wheel momentum-storage model would additionally show whether the wheels approach $h_{\max}$ (**??**) during this maneuver, which the current model cannot report because wheel momentum is not tracked as a state.
- **Single scenario, not a robustness study.** Sec. 6 reports one initial condition and one target attitude (Table 3). The Lyapunov result in Sec. 4.3 is almost-global and gives con-

fidence the controller would converge from other initial conditions, but settling time, peak torque, and saturation duration are all scenario-dependent quantities that have only been characterized at this one operating point.

- **Uniform scalar gains applied to a non-isotropic plant.** The gains in Table 2 were sized in Sec. 5.3 using the larger transverse inertia $J_x = J_y$; because J_z is roughly five times smaller, the same scalar K_p, K_d applied per Eq. (4.7) produce a different effective closed-loop bandwidth about $\hat{z}_B$ than about $\hat{x}_B, \hat{y}_B$ (Eq. (5.3)). This was an implicit consequence of the scalar-gain formulation, not a deliberately tuned design choice, and is worth flagging explicitly rather than leaving implicit.
- **No sensor or estimation subsystem.** Consistent with the scope defined in **??**, $\boldsymbol{q}$ and $\boldsymbol{\omega}$ are fed to the controller exactly; the results characterize the control law in isolation, not a complete ADCS.

7.3 Possible Improvements

The following extensions follow directly from the limitations above, ordered to roughly match their priority in Sec. 7.2:

1. **Add an environmental disturbance torque model** (at minimum, gravity-gradient torque, which requires no additional actuator and only depends on orbit geometry and the inertia tensor already computed in **??**) and evaluate steady-state pointing error under sustained disturbance, not just convergence from a torque-free initial condition.
2. **Model reaction wheel angular momentum as an explicit state**, coupling it into the total system angular momentum balance rather than treating commanded torque as a free external input, and add a magnetorquer-based momentum desaturation law – turning the idealized actuator of **??** into a physically complete one.
3. **Characterize robustness over a range of scenarios** (a sweep of initial tumble rates and target attitudes, and a parametric sweep over inertia tensor uncertainty relative to the nominal value in **??**) rather than the single case in Sec. 6.
4. **Move to per-axis (diagonal matrix) gains**, $\mathbf{K}_p, \mathbf{K}_d \in \mathbb{R}^{3\times3}$ sized individually from J_x, J_y, J_z via Eq. (5.3), to give uniform closed-loop bandwidth on all three axes rather than the axis-dependent bandwidth noted in Sec. 7.2.
5. **Add a sensor and attitude estimation subsystem** (e.g., a simulated gyroscope and star tracker or sun sensor feeding a multiplicative extended Kalman filter) so that closed-loop performance can be evaluated against realistic attitude knowledge error rather than the truth-model feedback used throughout this report.
6. **Extend to a redundant four-wheel pyramid configuration** with a pseudo-inverse control allocation law, replacing the trivial one-to-one torque mapping of Sec. 4.4 and directly addressing the single-point-of-failure limitation noted there.
7. **Replace the fixed inertial target with a time-varying, orbit-dependent pointing law** (e.g., nadir- or Sun-pointing via an LVLH reference frame), closing the scope simplifica-

tion identified in Sec. 2.1.

8 Conclusion

This report developed, from first principles, a quaternion-based attitude dynamics and control simulation for a 3U CubeSat and used it to close the loop on a specific, physically motivated maneuver: recovering from a post-deployment tumble of 19.7°/s while simultaneously executing a 120° large-angle slew to a target inertial attitude, using a three-axis reaction wheel assembly and a quaternion PD controller.

Three results, taken together, support the claim that the simulation is a correct implementation of the derived model, not merely a plausible-looking one. First, the closed-loop system converged to within 0.11° of the target attitude and 0.01°/s of zero rate in 74.1 s (Sec. 6), consistent in direction and rough magnitude with the settling time predicted by the small-angle gain design, given the additional gyroscopic coupling and brief actuator saturation the linear design explicitly did not account for (Sec. 7.1). Second, three independent numerical checks – quaternion norm conservation, agreement between two structurally different integrators, and monotonic decrease of the Lyapunov function derived in Sec. 4.3 – all held to within floating-point or negligible-physical-significance tolerances (Table 4). Third, and most tellingly, the Lyapunov monotonicity check was not merely a passive confirmation: it is what surfaced a genuine multiplication-order error in the first implementation of the error quaternion (Sec. 6.4), an error that produced superficially reasonable-looking convergence but violated an exact, unconditional analytical guarantee. Recovering from that error and re-verifying against the same check is, in the author's judgment, as informative a demonstration of understanding the underlying control theory as the final converged result itself.

The scope of this work was deliberately bounded, and that boundary is repeated here rather than left implicit. This report addressed the rigid-body attitude dynamics and control problem in a deterministic, disturbance-free, truth-model setting, with idealized actuation. It did not address attitude determination (sensor modeling and state estimation), environmental disturbance torque rejection, reaction wheel momentum management, or actuator redundancy – each identified explicitly in Sec. 7.2 as a specific, prioritized direction for extension rather than a vague caveat. Of these, disturbance rejection is judged the most consequential gap: a controller that has only been shown to converge in a torque-free environment has not yet been shown to hold a pointing requirement against the persistent, low-level torques a CubeSat actually experiences on orbit, which is the operational condition the ADCS exists to handle.

Within that bounded scope, the project met the objectives set out in **??**: the quaternion kinematics and Euler rigid-body dynamics were derived and explained physically, not just stated; a representative CubeSat mass/inertia model and actuator suite were defined with explicitly stated assumptions; a PD attitude controller was derived and proven almost-globally asymptotically stable; and the resulting closed-loop system was implemented, exercised on a large-angle maneuver chosen specifically to be representative of the class of problem the quaternion representation is preferred for, and validated against both numerical and analytical checks. The most direct extension of this work – and the one the author would prioritize first – is adding a disturbance torque model and re-evaluating the same controller's steady-state performance against it, since that is the single change most likely to alter the conclusions drawn here.

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